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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 12/30/2024 |
| **PI Name:** | Bell, Suzanne Ph.D. | | |
| **Project Title:** | Generalizability of Operational Feasibility and Acceptability of HFBP Exploration Measures in Nezemnyy Eksperimental'nyy Kompleks (NEK) | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders<br>(2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77058 | **Congressional District:** | 36 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 08/01/2018 | **End Date:** | 07/31/2024 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | October 2021 at time of Task Book entry: In 2018 Peter Roma was the Principal Investigator (PI), CoIs were Lauren Landon, Sandra Whitmire. In 2020 PIs were Suzanne Bell and Peter Roma and CoIs were Lauren Landon, Gloria Leon, Erin Flynn-Evans. In 2021 PI is Suzanne Bell (NASA), with CoIs: Lauren Landon (KBR), Gloria Leon (Independent Consultant), and Erin Flynn-Evans (NASA). March 2022 addition: Sheena Dev Ph.D. was added as Co-I in May 2021. May 2023 addition: Steven Anderson Ph.D. was added as a Co-I in October 2023. | | |
| **COI Name (Institution):** | Landon, Lauren Ph.D. ( KBR/NASA Johnson Space Center )<br>Leon, Gloria Ph.D. ( Independent Consultant )<br>Flynn-Evans, Erin Ph.D. ( NASA Ames Research Center )<br>Dev, Sheena ( KBR/NASA Johnson Space Center )<br>Anderson, Steven ( KBR ) | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

**Task Description:**

Given the small sample sizes associated with space and analog research, it is critical that comparable data using the same set of measures on key Behavioral Health and Performance (BHP) variables are collected across participants.
The purpose of this task is to establish, evaluate, and manage a common or “standard measures,” called Human Factors Behavioral Performance -- Exploration Measures (HFBP-EM) for use in spaceflight and analog research. These measures will be used to develop baselines, systematically characterize risk likelihood and consequences, and assess effectiveness of countermeasures for human factors and behavioral performance risk factors. This task allows these measures to be consistently implemented with evidence-based iterations over time, allows comparability with other HFBP-EM applications, and data sharing with Nezemnyy Eksperimental’nyy Kompleks (NEK) investigators as needed to inform Human Research Program (HRP)’s research gaps.

The proposed research is a supplement to the previous directed task of developing and evaluating a standardized suite of measures that can capture space-relevant behavioral health and performance phenomena across spaceflight and analog environments (called HFBP Exploration Measures or HFBP-EM, formerly Behavior Health and Performance [BHP] Standard Measures). The current proposal includes HFBP-EM in the Scientific International Research In a Unique terrestrial Station-21 (SIRIUS) mission and augments the protocol with a measure of habitability and Russian language personality, affective, and group dynamics measures. To date, most HFBP-EM data have been collected in the Human Exploration Research Analog (HERA) at Johnson Space Center. These missions are up to 45 days and include US participants. Data collection in the SIRIUS missions provides an important data collection opportunity for a longer-duration mission (i.e., 8 month), in a high fidelity mission simulation analog (i.e., NEK) with habitability features different from HERA, and from a multinational crew.

The differences in habitat (i.e., NEK versus HERA) allow for testing of the Spaceflight Habitability Acceptability Questionnaire (SHAQ). SHAQ measures the BHP impacts of net habitable volume and layout, which allows for additional risk coverage (Human Research Roadmap gap HSIA 101). SHAQ will allow for comprehensive and systematic evaluation of existing operational habitats (e.g., International Space Station, spaceflight analogs), habitats in development (e.g., Orion, Gateway), and multiple prototypes across mission types (e.g., lunar surface, Mars transit, Mars surface) to inform habitat designs, recommendations, and requirements.

Our research extends previous HFBP-EM efforts by including Russian translated measures of personality, affect, and group dynamics that have been collected previously in analog environments. This will allow us to further examine the feasibility and acceptability of the HFBP-EM suite, as compared to other measures taken in the preferred language of the participant. It will also increase the scientific yield in NEK.

Aims:

1. Provide a set of BHP standard measurements for investigators to use in proposed projects.

2. Enable comparison of multiple missions across spaceflight analog campaigns to quantify risk using reliable metric-based data.

3. Provide database for data-mining and integrative modeling and increase research data quality and transfer to Life Sciences Data Archive (LSDA).

Specific Aims:

The specific aims of our research are to:

Aim 1: Assess Operational Feasibility of HFBP-EM in the Russian NEK space analog.

Aim 2: Assess Operational Acceptability of HFBP-EM in the Russian NEK space analog.

Aim 3: Provide a set of HFBP-EM for data-sharing with intramural and extramural investigators to use in proposed projects.

Aim 4: Enable comparisons of multiple missions across spaceflight analog campaigns to quantify risk using reliable metric-based data.

Aim 5: Contribute to HFBP-EM database for data-mining and integrative modeling and increase research data quality and transfer to LSDA.

Aim 6: Provide an assessment of behavioral and health performance impacts of habitat layout and acceptability.

Aim 7: Provide a comparative assessment of individual and crew functioning in isolation with measures taken in the participants’ preferred language with those collected in English.

Aim 8: Examine the extent to which a monthly retrospective sleep quality index relates to daily subjective and objective sleep measures, as well as subjective ratings of neurobehavioral functioning and mood over time.

This task will contribute to gap closure by facilitating all tasks using the Institute of Biomedical Problems (IBMP) NEK Chamber analog in the risk areas of "Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders" (BMed), "Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload" (Sleep), and "Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team" (Team).

Research Methods:

Data will be collected from the Ground-Based Experiment Complex (NEK) space analog facility at the Russian Institute for Biomedical Problems (IBMP). Pre-mission data will be collected from all individuals (primary crew and backups).

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| **Rationale for HRP Directed Research:** | This proposal qualifies for a directed study due to the urgent, time-sensitive need to provide “standard measures” as the foundation to achieve consistent research measures for data-sharing in NEK and to meet the highly constrained, operationally-focused data gathering and analysis that allows for greater consistency in the research methods that are very specific to NASA Human Research Program (HRP) standard measures development. Timelines for integration of study hardware and protocol are further constrained due to international agreements, Export Control, and other requirements that modify/accelerate schedules. Completion of the required research, support to both analog and operational requirements, and the required vetting of the evidence-based standards makes the solicitation process prohibitive. Moreover, the projected start date of the first SIRIUS study is rapidly approaching.<br>In 2020: The proposed research is a supplement to the previous directed task of developing and evaluating a standardized suite of measures that can capture space-relevant behavioral health and performance phenomena across spaceflight and analog environments (called HFBP Exploration Measures or HFBP-EM, formerly Behavior Health and Performance [BHP] Standard Measures). The current proposal includes HFBP-EM in the SIRIUS-21 mission and augments the protocol with a measure of habitability and Russian language personality, affective, and group dynamics measures. This project qualifies for a directed task because of: (a) the need for NASA to ensure continuity across all HFBP-EM data collection; (b) the importance of providing data to other internal and external researchers in a timely manner; and (c) the ability for the lab to report quickly on data collection success and barriers. Further, the time sensitive and urgent need to develop a validated standard set of measures that can be used across laboratory, analogs, field, and spaceflight, further qualifies this research as a directed study. |
| **Research Impact/Earth Benefits:** | HFBP Exploration Measures provides a comprehensive understanding of the behavioral medicine and team dynamics in isolated and confined settings. Increasingly, the American population is isolated from friends, family, and coworkers due to quarantine and remote work. In addition, there are special populations such as prisoners and nursing home residents who can encounter extended separation from family and friends. Our research provides insights into the psychological impact of isolation and confinement as well as how the circumstance can shape relational and team dynamics over time. |
| **Task Progress:** | This entry reports on task progress from 6/2/2023 to 6/2/2024. During this time, the Behavioral Health & Performance (BHP) Laboratory completed all aspects of the Human Factors Behavioral Performance – Exploration Measures (HFBP-EM) data collection in the NEK SIRIUS21 8-month mission, including data cleaning, processing, and scoring of all HFBP-EM data collected. A final summary report has been written and submitted to the NASA Human Factors and Behavioral Performance (HFBP) element. The primary purpose of this final report was to summarize HFBP-EM results from NEK SIRIUS21 and make recommendations for future long-duration missions using the HFBP-EM suite. The report includes pre-mission, in-mission, and post-mission data for the entire SIRIUS21 study. In addition to addressing the predefined specific aims of this study, the final report also provides an examination of the impacts of the unexpected egress of a crewmember on mission day 32 on behavioral health and team functioning. Please review the “Human Factors and Behavioral Performance Exploration Measures in NEK SIRIUS21: Final Summary Report” (PI: Bell) for additional details. All HFBP-EM data shares for SIRIUS21 have been delivered to collaborating researchers and final data has been transferred to NASA’s Life Sciences Data Archive (LSDA). |
| **Bibliography Type:** | Description: (Last Updated: 08/01/2025) |
| **Abstracts for Journals and Proceedings** | Anderson SR, Dev SI, Bell ST. "The development and application of cognitive performance assessment for exploration class mission EVAs." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024.<br>Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 |
| **Abstracts for Journals and Proceedings** | Bell ST, Anderson SR, Landon LB, Dev S, Glaros Z, Keller B, Flynn-Evans E. "Human factors and behavioral performance exploration measures in NEK. " 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024.<br>Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 |
| **Abstracts for Journals and Proceedings** | Leon GR, Keller BA, Miller JCW, Bell ST. "Personality, affect, and coping among SIRIUS 21 crew." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024.<br>Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 |
| **Articles in Peer-reviewed Journals** | Bell ST, Anderson SR, Roma PG, Landon LB, Dev SI. "Social support from different sources and its relationship with stress in spaceflight analog environments." Front Psychol. 2025 Jan 15;15:1350630. https://doi.org/10.3389/fpsyg.2024.1350630 , Jan-2025 |
| **Articles in Peer-reviewed Journals** | Landon LB, Miller JCW, Bell ST, Roma PG. "When people start getting real: The group living skills survey for extreme work environments." Front Psychol. 2024 Apr 15;15:1348119. https://doi.org/10.3389/fpsyg.2024.1348119 ; PubMed PMID: 8689722; PubMed Central PMCID: PMC11060178 , Apr-2024 |